

Nathan Svoboda

630-532-9604 | njsvoboda04@gmail.com | Woodridge, Illinois | [ennjaywithagreyhat.github.io/portfolio](https://github.com/EnnJayWithAGreyHat) | github.com/EnnJayWithAGreyHat

Education

Bachelor of Science, Computer Science at *Barrett the Honors College at Arizona State University* August 2022 - May 2026
- Awards: Dean's List 5 Semesters

Technical Skills

Programming Languages: Python, JavaScript, C, C++, Java, SQL, Bash

Technical Proficiencies: IDA, BinaryNinja, GDB, Pwntools, x86, Yan85, Wireshark, Nmap, Burpsuite

Project Management: Git, Scrum, ClickUp

Professional Experience

Full-Stack Software Engineering Intern - Hawkeye Energy Solutions May 2024 - Present

- Implemented React UI components and state management to enable customers to configure device thresholds, satisfied customer request and improved configuration speed for operators by 53%
- Redesigned the backend of a React application to fix API call and data resolution errors by fixing Java methods that would communicate directly with building meters through our integrated servers
- Spearheaded the migration and integration of 120,000 historical metering data points from an older database to a newer and more dynamic database using SQL to provide building metering trends from past years
- Worked in a CI/CD environment for multiple customer-facing applications utilizing npm, Gradle, GitHub, Java, and the Niagara framework to maintain and develop applications and documentation for customers

Software Engineer - Paragon Autonomous January 2024 - May 2025

- Led software development for autonomous fire surveillance drone and implemented A* pathfinding in C++ using point clouds, enabling autonomous navigation with tests
- Researched and developed a template framework in C++ to simplify code deployment with flight controllers, and ensured compatibility with Mavros, Gazebo, and Ardupilot for drone programming and functionality
- Tested the Lidar and Ultrasonic sensors to see the information that is provided, and how that data can be integrated into our code for adaptation of flight controls, and collision avoidance

Projects and Involvement

Yellow Belt and Coursework | *GDB, IDA, BinaryNinja, Pwntools* August 2024 - Present

- Leveraged pwn.college challenges to develop practical skills in exploiting vulnerabilities and analyzing binaries using IDA, GDB (gef), Python, Bash, and other tools to earn ctf flags and earn my yellow belt
- Crafted and executed buffer overflow, tcache corruption, ROP, Yan85, shellcode injection (with ASLR and NX), DHKE handshakes, SHA collisions, and POA exploits on binaries to simulate attacks on vulnerable programs
- Used example web servers to practice exploits relating to web security, SQL injections, HTTP authentication bypassing, command injection, XSS, CSRF, cookie manipulation, ARP spoofing, and MITM attacks

3rd Place Cactus Con CTF 2026 | *Burpsuite, Python* February 2026

- Found a web vulnerability related to unsafe image rendering, and utilized a CSRF attack with the Burpsuite repeater to send commands to a remote server
- Worked on a flag regarding utilizing an error with the JWT RSA signature that allowed us to use our signature to sign the JWT with a known key, decrypt the admin cookie, get an admin session, and a flag
- Expanded my knowledge regarding scripting, privilege escalation, binary exploitation, and reverse engineering techniques through various other challenges

Offensive Cybersecurity Involvement at ASU | *GDB, Python, GoLang* August 2024 - May 2026

- Participated in L3akCTF, Cyber Apocalypse CTF, and b0lersCTF with CTF Academy, solving challenges across various fields from bypassing LLM guards via prompt injection, to efficient coding, and cloud exploitation
- Exploited vulnerabilities in 10MB+ games such Tasteless Shores written in GoLang, and CSGO, written in C++ to trace live code, identify vulnerable game logic code, exploit physics and RNG, and uploading code patches
- Participated in Hack the Madness 2026 which entailed brute forcing passwords with Hydra, database exploits exploring 9 rows with sqlmap, binary exploitation with ROP, digital forensics with Autopsy, and OSINT